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Successful Implementation and Performance of Surfactant Polymer Flood Pilot in Mehsana Asset

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Abstract

The southern sector of the mature oil field N has a primary recovery of around 42.6% due to relatively favorable reservoir conditions (high pay thickness, high permeability, active edge water drive, moderate viscosities of 10-25 cP) in K-I sand. Therefore, laboratory studies recommended a tertiary recovery method using Surfactant Polymer (SP) Flooding for altering the subsurface reservoir/oil parameters to reduce the residual oil saturation and improve the displacement and sweep efficiency. A pilot project comprising of a normal five spot pattern with 6 months surfactant slug and 2 ½ years of chase polymer slug is under progress since March 2023. The objectives of the pilot study are to test the efficacy of the process in active water drive, gain operating experience, establish monitoring programs and evaluate techno-economics for field implementation.

The SP pilot was implemented by converting four existing producers into injectors at the structural updip part of the block. The monitoring approach is divided into two programs: the injection plan and the production plan. The injection plan consisted of key parameters such as injection rate, injection head pressure, viscosity of the slug, surfactant concentration, polymer concentration, co-solvent concentration and total salinity. These were based on laboratory and core flooding studies done to achieve continuous microemulsion phase. Other parameters like dissolved oxygen, iron, turbidity, hardness, oil and grease content, pH, filter ratio, SRB, etc., are measured for injection quality control. This injection plan is one of the key reasons for the success of the chemical flood.

The production plan consisted of the pattern and offset producers' response recorded in terms of changes in water-cut, produced chemical slug concentrations, liquid withdrawal, and oil rates and thereby mapping of the movement of the chemical slug. Time-lapse saturation monitoring in injectors and producers, and changes in oil and water composition with flooding are also monitored.

The direct producer witnessed a water cut decline from 80%-60%, and the oil rate improved by over 600% (3 m³/d to 20 m³/d). Two of the offset wells also showed improvement in terms of oil rate during the flooding process (85-300% increase in oil rate). The pilot oil rate performance is in accordance with the planned profile, while the decline in water-cut is more significant than the plan. Comprehensive monitoring and quality control procedures are followed to ensure smooth operations. Pilot performance

demonstrated the process's effectiveness within two years for K-I sand which has high pay thickness and strong aquifer drive. During the process, challenges encountered were in regard to downhole equipment failures, loss of productivity, measurement methodology of produced chemical concentration and chemical quality assurance. This paper introduces the field with its historical production and then deliberates the operational aspects, quality control, monitoring program, performance analysis and challenges faced in SP flooding.

Introduction

China has given the world many ASP flood projects. From 1994 to 2000, 8 ASP pilot tests were conducted in the high permeable sandstone formation of Daqing field with average pilot area oil production rate increased from 36.7 to 91.5 m³/d, while water cut decreased from 82.7 to 59.7%. Incremental recovery efficiencies from the five completed projects varied from 19 to 25% OOIP. Shengli oil field reservoir is a channel-sand deposit with darcy level permeability with the first ASP pilot started in 1992 and completed in 2002. The oil rate increased from 100 to 237 m³/d at peak production, and corresponding water cuts decreased from 96 to 83% (Olajire, 2014). The total incremental recovery was 15.5% OOIP. In 2003, 13 field cases were carried out in high permeability sandstones and conglomerates, having high temperature and high salinities. Shengli oil field had an improved recovery of 12.1% over water flood in SP pilot and 10.3% in expanded SP flood (Thomas, 2019). There are ASP cases in Canadian oil fields as well. The mature Taber South field of medium gravity oil produced 48 m³/d before the ASP flood in June 2006. Production increased to 239 m³/d as of December 2007 with oil cut rise from 1.7 to 7.3% over the same period. Taber Glauconitic Pool oil production was 26 m³/d with an oil cut of 0.9% when ASP injection began in January 2008 and in April 2010 oil production enhanced to 127 m³/d with 5.3% oil cut. Total incremental oil production above the base decline at the end of April 2010 amounting to 0.4% of OOIP as incremental recovery factor. The Phase One ASP scheme of Mooney, Canada witnessed production increase from the ASP flood with base oil rate of 28 m³/d in 2011 to 413 m³/d in December 2013. (Watson, et al., 2014). The earliest field testing of ASP flooding in the USA was at West Kiehl, Wyoming in September 1987. The incremental oil recovery over the next 2.5 years had reached 26% OOIP. The next ASP pilot test came from the Cambridge Minnesota field with an increased recovery of 28% of OOIP. Tanner Field ASP had an incremental OOIP of 17%. An ASP flood in the Sho-Vel-Tum field was conducted beginning in 1998. The ASP project increased average oil recovery from 0.6 m³/d to 3 m³/d and added 1660 m³ of incremental oil over a period of 1.3 years. In 2008-pilot ASP injection started in Lawrence field Illinois and the performance indicated by an increase in the oil cut ratio in the pilot wells (Olajire, 2014). ASP for a clastic field in Southern Sultanate of Oman recovered significant incremental oil (20%) post water-flood with 31% reduction in oil saturation and significant water-cut reversal (lowest upto 25-30%) (Al-Shuaili, et al., 2018).

There are three ASP/SP case studies in India. Firstly, Viraj field which is a sandstone reservoir with average porosity of 30% and permeability ranges of 1-10 Darcy. The reservoir crude has viscosity of 50 cP at reservoir temperature of 81°C. ASP with four inverted 5-spot patterns had been commissioned on 10th August-2002. Before implementation the ASP pilot in July'2002, oil rate and water cut were 152 m³/d and 89% and after implementation of ASP pilot, water cut declined to 72% by April'2003 and oil production increased to 260 m³/d, indicating the efficiency of ASP flooding. Thereafter the water-cut increased to >80% with time. Midcourse correction led to drilling of additional wells to capture the mobile oil which increased oil production from 100 m³/d in March'2010 to a level of 160 m³/d whereas water cut has remained at around 83% (Singh, et al., 2017). Then, there is Jhalora field, a sandstone reservoir with darcy level permeability and oil viscosity of 30-50 cP at reservoir temperature. One inverted 5-spot ASP pilot was designed in February 2010. The pilot indicated a decline in water-cut and increasing oil rate. Till October 2011, the incremental oil was 7472 m³ (Jain, et al., 2012). Lastly, there is Mangala Field with multi-layered sandstone with darcy level permeability and paraffinic oil with average viscosity of ~15 cP and wax appearance temperature only

about 5°C lower than reservoir temperature of 65°C. ASP pilot was implemented with a normal five spot pattern in April 2014. The oil cut of the central producer increased from 10% to 80%. The oil production rate showed almost an eight-fold increase from 8 m³/d to nearly 64 m³/d. The incremental oil recovered in the ASP pilot was 3975 m³, which is greater than 20% of the pilot area STOIP. The remaining oil saturation is less than 5-10% in the ASP area (Pandey, et al., 2016). In the subsequent sections, the study area is introduced with rock–fluid characterization and historical production performance. This is followed by a detailed discussion of the pilot plan, surface facility design, and the monitoring program adopted.

Study Area

N Field can be divided into two sectors based on oil viscosity under reservoir conditions as shown in Figure 1. Northern sector has oil viscosity in the range 50-120 cP, while southern sector has viscosity in range 15-22 cP. These reservoirs are operating under very strong aquifer support. Current recovery for northern sector is 26.2 % while from southern sector it is 42.6 %. Also, despite having higher oil recovery, water cut in southern sector (76 %) is lesser than northern sector (88 %). This difference in performance arises from viscosity contrast in these sectors. Polymer flooding scheme is ongoing in the Northern Sector of Field since 2020 in 11 inverted five spot pattern with incremental oil of ~170 m³/d and based on the encouraging performance field expansion with 46 additional patterns is planned.

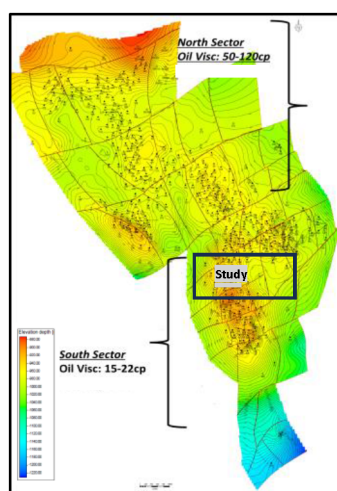


Figure 1—Field N Relief Map of K-I sand marked with the study area and viscosity gradation.

The southern sector consists of multiple structural highs with four well-defined reservoir sands—K-I, K-IB, K-II, and K-III—from top to bottom. These channel sands exhibit excellent reservoir characteristics, with porosity around 25% and permeability ranging from 2 to 5 Darcies. The study area is confined to a block in the southern sector. The study focuses on K-I sand as 65% of STOIP of the southern sector is in K-I sand, and K-I sand contributes 75% of southern sector oil production. The log correlation of the wells in the study area is shown in Figure 2.

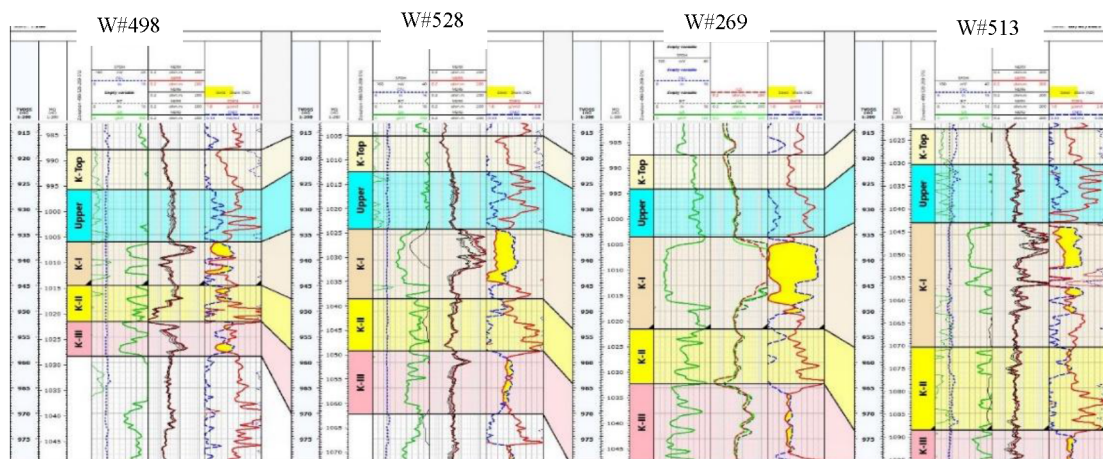


Figure 2—Log Correlation in the Study Area (from North to South) with K-I, K-II, K-III sands.

Block History

This block has GOC at 948 m determined from the logs of W#174 drilled in 1984 and OWC at 988 m as shown in the relief map of the area (Figure 3). The block's production history begun with the drilling of W-37, W-61, W-69, W-70, W-71 and W-77 from the oil rim. Subsequently production from the gas cap region started from W-30, W-36, W-54 W-169, W-176, W-203, W-217, W-269, W-326, W-341 and W-382. This led to gas blow down and movement of oil in the gas cap region as observed in the wells drilled updip W-542, W-528, W-513 etc. In addition to that OWC movement was also observed to the depth of 948 m (as per last drilling in 2015-16). The performance plot of the block is placed at Figure 4. From 1972 to 1986, production data has a step like pattern as the data was allocated on annual basis. For the next two decades, the oil productivity remained nearly constant to 15 m³/d per well. Thereafter, oil rate declined to 4-6 m³/d in-spite of increasing the flowing strings with in-fill drilling and water conformance control due to rise in water cut and gas blow down. The decline rate was 8% as on 01.04.2023. The major oil producers historically in this region were W#36, W#37, W#70, W#71, W#77, W#265, and W#432 with cumulative oil greater than 50,000 m³ per well. At the onset of the pilot project, there were 15 flowing wells contributing 55 m³/d of oil with W-432, W-217 and W-269 production more than 5 m³/d oil rate (refer Figure 5).

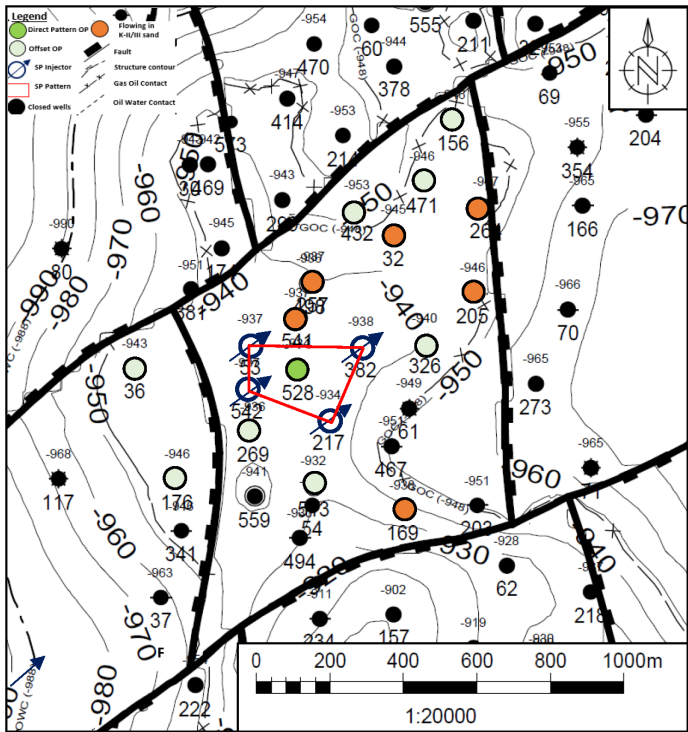


Figure 3—Part Relief Map of the study area (Zoomed in) with marked normal five spot pattern.

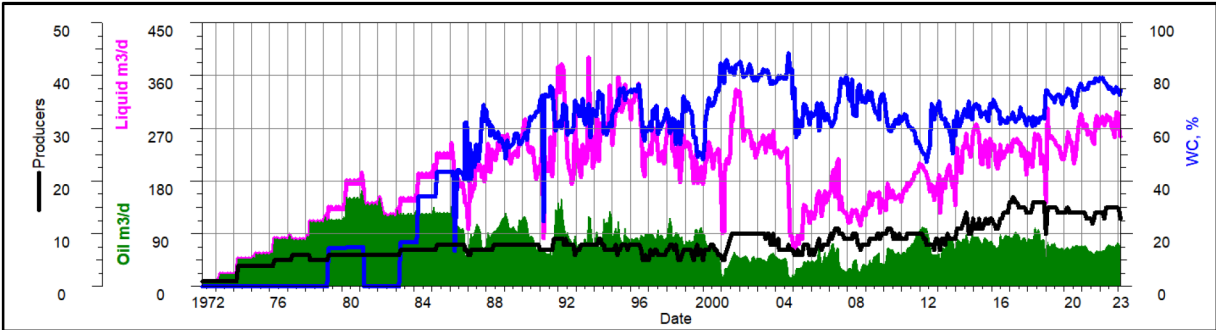


Figure 4—Performance Plot of the Study area since inception till March 2023.

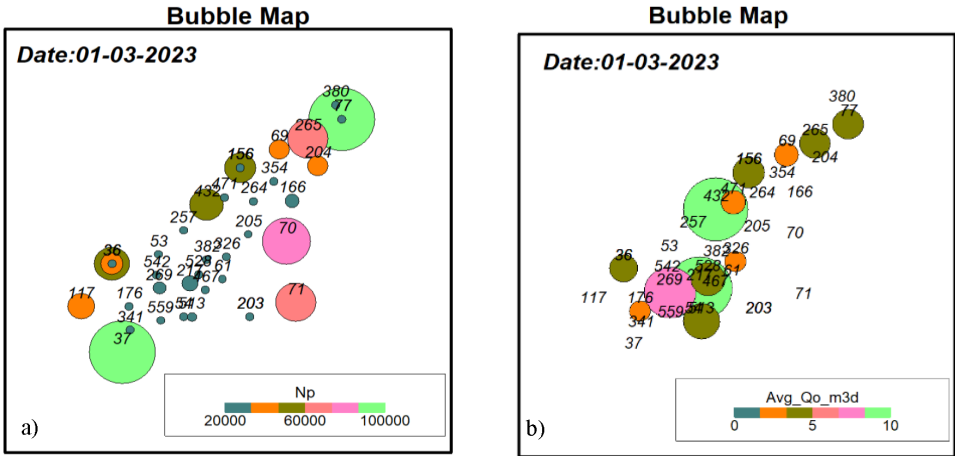


Figure 5—Bubble Map of the Study area pre-flood a) Cumulative Production b) Average Oil rate, m³/d

Pilot Plan

The following were the results of the laboratory and numerical simulation carried out by Institute of Reservoir Studies (IRS), ONGC Ahmedabad:

1. The pilot area selection was based on high remaining/residual oil reserves, minimal geological heterogeneity, high pay thickness, optimal inter-well spacing, and limited water cut variation.
2. A normal five spot pattern with existing wells conversion potential to surfactant-polymer injectors W-53, W-382, W-217 and W-542 and direct oil producer W-528, as shown in [Figure 3](#).
3. Production rate of 65 m³/d and injection rate of 65 m³/d/well was determined with 0.3 PV of Surfactant-Polymer Slug for 0.5 years and 0.7 PV of Polymer Slug for 2.5 years.
4. Total oil of 12,210 m³ from the direct pilot producer due to SP flooding.
5. Surfactant concentration: 3000 ppm, Polymer concentration: 1000 ppm, 1% Co-solvent and 1% NaCl.

The key parameters of the pilot are mentioned in [Table 1](#):

Table 1—Key Parameters of Pilot Plan

Category	Value
Pattern Type	A normal five spot pattern
Injectors	4 (542, 382, 217, 53)
Direct Producers	1 (528)
First line offset producers	5 (269, 326, 36, 513, 432)
Second line offset producers	2 (176, 471)
Injector-Producer distance	132-183 m
Injection rate	260 m ³ /d in 4 wells.
Production rate	65 m ³ /d
Pilot Pore Volume	0.124 MMm ³
Pilot Target Oil	0.036 MMm ³

Surface Facilities Design

The injection process begins with feed water supply from the Effluent Treatment Plant. This water undergoes a series of treatments to ensure its suitability for injection solution preparation. First, contaminants such as oil, grease, bacteria, iron, dissolved oxygen, and total suspended solids (TSS) are removed through filtration and chemical dosing. A biocide (Aldehyde and Amine based) is added to prevent bacterial growth, while an oxygen scavenger (Sodium Sulphite) eliminates dissolved oxygen, ensuring the water is free from corrosive elements. The treated water is then stored in nitrogen-blanketed tanks to prevent air contamination. The schematics of the injection plant is placed as [Figure 6](#).

In the chemical preparation units, surfactant, polymer, co-solvent, and salt solutions are prepared separately. Surfactant and co-solvent are unloaded and stored in buffer tanks with dedicated metering pumps. Polymer, supplied in dry form, is dissolved using a Polymer Slicing Unit (PSU) and mixed thoroughly in a maturation tank to create a homogeneous solution. Similarly, salt is dissolved in preparation tanks and integrated into the solution. All the solution preparation process happens in a nitrogen blanketing environment so that no oxygen ingress takes place in the system preventing the degradation of polymer/SP slug.

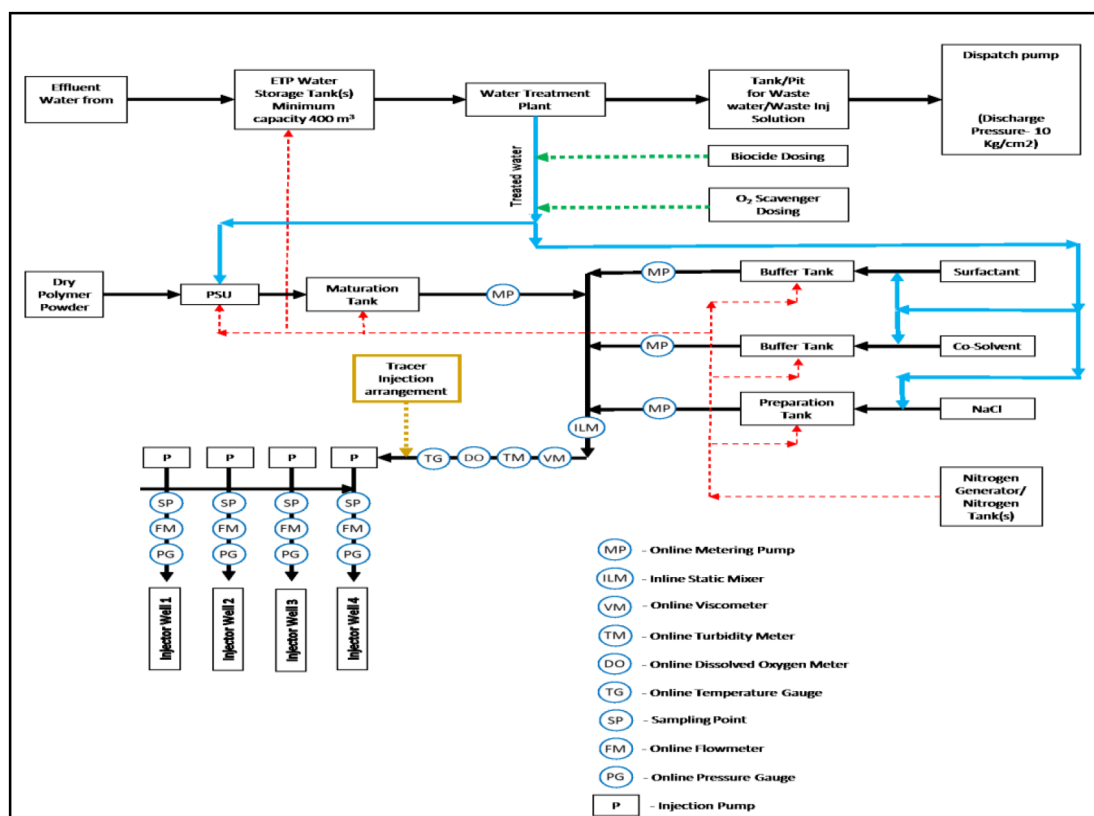


Figure 6—Schematics of the Surfactant Polymer plant Injection Facility.

The individual chemical components are then combined in a static mixer, where they form the surfactant-polymer solution (SP slug) or the polymer solution (polymer slug). The SP slug, used during the first six months, is formulated with surfactant (3000 ppm), polymer (1000 ppm), co-solvent (1%), and salt (1%). The polymer slug, used subsequently for next 30 months, has a higher viscosity (7.2-9.0 cP at 73°C) to chase the SP slug.

The quality of the prepared solution is constantly monitored using online instruments, including viscometers for viscosity, dissolved oxygen meters, and flowmeters. Manual sampling points also ensure critical parameters like filtration ratio, turbidity, and chemical concentrations are within specified limits.

After the required SP/P slug is prepared and stored, it goes through high-pressure injection pumps equipped with variable frequency drives (VFDs) which deliver the prepared solution to the injection wells. Each of the four designated wells receives the solution at a precise rate of 65 m³/day. Finally, the solution is injected into the reservoir, reducing the adverse mobility ratio and enhancing oil recovery in the targeted zone.

Monitoring Program

The monitoring program (designed by IRS, Ahmedabad) is divided into two sections, viz. Injection and Production for the pilot project. In the injection monitoring program, parameters monitored for water treatment, quality assurance of chemicals and chemical slug properties are included. The production monitoring program comprised of well monitoring techniques viz. well testing, water-cut sampling, polymer and surfactant concentration and liquid level measurement, oil and water analysis using SARA, viscosity and ionic composition, and saturation monitoring by logging (using sigma, C/O and SATG measurements).

Injection

- **Water Treatment:** Multiple level measurements of injection fluid parameter are required to inject desired slugs successfully. The Feed water for the project is available from the nearest Effluent treatment plant. Feed water quality is measured for basic parameters like TSS, Iron, Salinity, Hardness, Turbidity, pH, Oil& Grease etc. Raw feed water gets treated by removal of TSS, Iron, SRB, Hardness and Turbidity to achieve the desirable parameter for injection solution preparation. The tolerance limit of these parameters is Hardness 200 ppm, Turbidity 10 NTU, Salinity 0.9-0.95% (as NaCl), pH:6.5-8.5, Oil& Grease 10 ppm, TDS 1.2%, and Iron 0.5 ppm. SRB nil counts/ml. These parameters are stringently measured in the treated and feed water properties as placed in the [Figure 7](#).

The effluent water quality is improved as the process progresses ahead and therefore turbidity, TSS and oil & grease remains below the tolerance limit even in the feed water. Anionic surfactants form nonactive complexes with divalent, resulting into overall IFT increase. Also, in presence of divalent or multivalent ions, surfactant stability at high salinity is much lower and it can precipitate into these complexes. The precipitation mechanism is complex, and these divalent/multivalent ions are consumed with time ([Sheng, 2011](#)). Polymer hydrolysis also increases with thermal maturation in the reservoir. With increase in hydrolysis, its affinity for divalent/multivalent ions increases resulting into polymer crosslinking and precipitation. Therefore, presence of hardness and iron is maintained as low as possible ([Zagitov, et al., 2020](#)).

Surfactant is an optimal salinity governed process; therefore, salinity of the feed water and soft water should be in line with the formation salinities. Polymer stability is also better in presence of sodium ions as they are least reactive to hydrolysed polymers. Sulfate based surfactants have limited stability at low pH and very high pH. Extreme pH conditions can lead to tight emulsions, silicate scaling tendency, therefore optimum pH is also maintained in the process ([Pandey, et al., 2016](#)). Finally high turbidity injection leads to reservoir damage, therefore tolerance limit on turbidity helps with continuous chemical slug injection.

- **Chemical Slug:** The four main parameters monitored for chemical slug are: the surfactant concentration (~3000 ppm), the cosolvent concentration (0.95-1.05 % of the surfactant concentration), salinity (1.9-2%), viscosity of both Polymer and SP slug (5.5-10 cP), filter ratio (<1.35 with 5-micron filtre) and dissolved oxygen (<0.08 ppm). The chemical slug properties for the entire duration of the project is placed as [Figure 8](#).

The right amount of surfactant concentration helps in attaining critical micelle concentration (CMC). CMC is the concentration of surfactants above which micelles are spontaneously formed. Below CMC, IFT changes with the increase in surfactant concentration and above CMC increasing surfactant concentration has minimal impact on IFT. Therefore, IFT is a direct function of surfactant concentration. Co-solvent prevents phase separation between polymer and affects the effective salinity for Type-III microemulsion as evident in the laboratory study. Co-solvent concentrations are stringently maintained in the system. Surfactant concentrations, IFT reduction and Optimum salinity are codependent on each other. Optimum salinity is the salinity where bi-continuous/Winsor Type-III microemulsion/Middle-phase emulsion is formed. This is the state where IFT reduces below 0.01 mN/m ([Sheng, 2011](#)).

Polymer in the surfactant slug avoids water fingering, while the polymer slug post SP improves the sweep of the surfactant mobilized oil. Polymer is mainly used for mobility control by increasing the water viscosity. So, viscosity of the SP slug is maintained between 5-10 cP in order to maintain unit mobility ratio in the reservoir. Dissolved oxygen is measured and kept below 0.08 ppm to avoid oxygen ingress in the polymer system as it reduces the polymer viscosity by chain

scission. Lastly, the compatibility of polymer solution and surfactant-polymer slug is established by measuring filter ratio just before injection to ensure flowability and injectivity of the slug.

- **QA/QC of Chemicals:** Polymer, Surfactant, and Co-solvent have corporate specifications where they are tested for quality assurance on a random sampling basis. Currently, the certificate of analysis from manufacturers is used prior to injecting the solution. The certificate of analysis covers all the required parameters of corporate specifications.

Surfactant specifications comprise of active matter (minimum 50%), physical state (homogenous), aqueous solubility (<3 NTU turbidity), pH (6-8), thermal stability (>90% after 7 days), CMC, IFT, emulsion formation and compatibility study with divalent ions. Co-solvent specifications contain purity (>99%), water content (<0.1%), platinum-cobalt color scale (10 maximum), appearance (clear), solubility (<3 NTU turbidity in aqueous form), boiling range (228-232 °C), density (~0.95 g/cm³) and refractive index (~1.43).

Polymer specifications are appearance (white powder), moisture content (<10%), apparent viscosity (32-45 dl/g), solution properties like relative viscosity, Brooke field viscosity, screen factor in presence of divalent ions, insoluble residue (<0.3% by mass) and thermal stability (80% viscosity retention for 7 days).

One of the major challenges of the project is that the corporate specifications take over two weeks of analysis and data compilation, and lot sizing is also of two weeks. This leads to onset of injection before the results of QC. Therefore, in the upcoming project, chemical storage will be kept for at least 3 months and molecular level fingerprinting of chemicals using Fourier transform infrared spectroscopy will be carried out for polymers and surfactants to expedite the QC process.

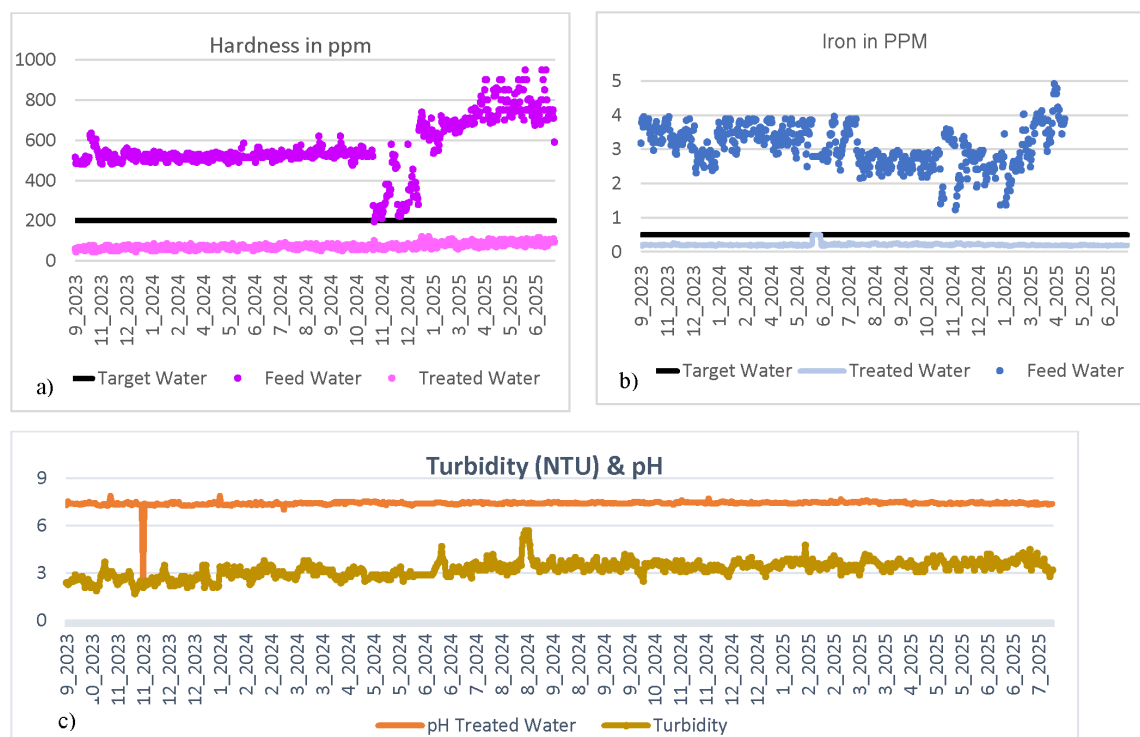


Figure 7—Water Treatment Parameter trend in Surfactant Polymer Plant a) Iron b) Hardness C) pH and Turbidity.

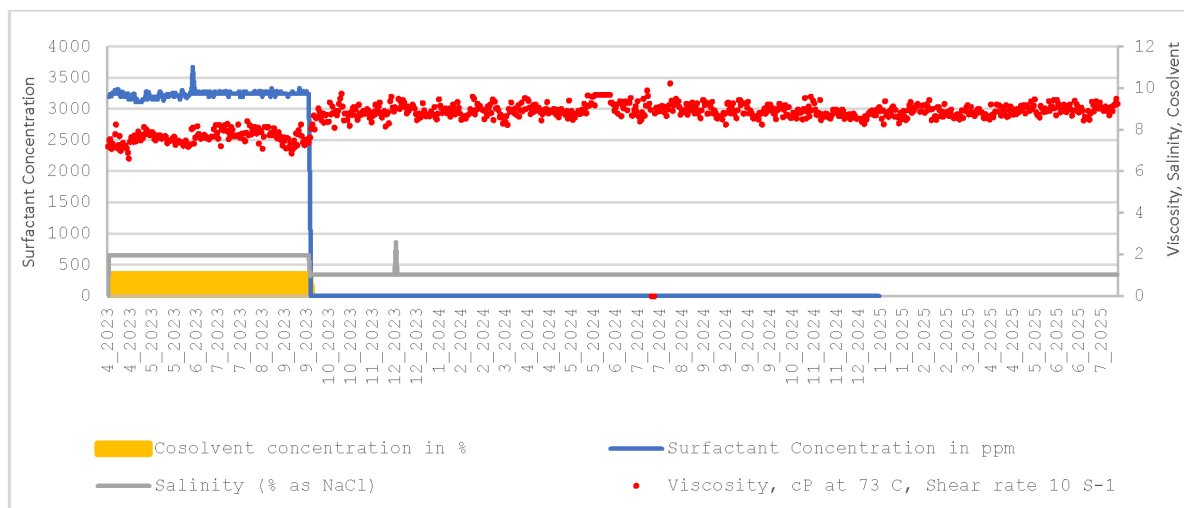


Figure 8—Injection Chemical Slug Properties: Concentrations, Salinity, and Viscosity in Surfactant Polymer Plant.

Production

- **Well Monitoring:** The direct producer of the pilot producer W#528 is tested 56 times with mass flow meter (two testing every month), well head water-cut samples are taken 10 times in a month and the pump liquid levels are measured once every month in 22 months of pilot operation. Well-head water-cut samples assist in validation of the results of the mass flow meter based on well testing. As the producer is on Progressive Cavity pumping, it's crucial to know the operating pump parameters for well optimization. The well testing plot of W#528 is shown in Figure 9. In addition to that, 7 offset wells were also monitored monthly with well flow measurements, well-head water-cut trends and liquid levels. Polymer and Surfactant concentrations are also monitored in the direct and offset producers for understanding the movement of the slug.
- **Oil & Water Analysis:** Mobilized oil is produced with SP flooding, changes in the oil composition and properties pre-flooding and during flooding appear imperative. The increase in aromatic composition indicates displacement of oil particularly the lighter fractions by reduction of IFT. SP flooding can change the resin-asphaltene equilibrium and the reduction in resin indicates the possibility of resin interaction with surfactants as shown in Figure 10. This can lead to asphaltene aggregation and precipitation (Alshaikh, et al., 2019). Oil analysis pre-flooding showed dry oil viscosity of 41 cP and during flood oil viscosity of 57 cP is measured at reservoir temperature. Water analysis indicates an increase in iron and sulphate ions which on interaction with hydrolyzed polymer increases the likelihood of formation of complexes. Decrease in calcium and magnesium concentrations demonstrate the stripping of divalent ions from dissolved state and formation of scales. Increase in TDS indicates an increase in the production of polymer as indicated in many fields scale polymer flooding due to polymer degradation and reservoir-rock interactions (Shankar, et al., 2022; Chang, et al., 2024) as shown in Table 2.
- **Saturation Monitoring:** Time lapse saturation logging snapshots are acquired in Injectors and Producers at the beginning of the SP flood and during the SP flood (Post the end of SP slug). In W#217, saturation log couldn't be recorded due to operational issues. The de-saturation of oil in the injectors W#53 (50% to <10%), W#382 (40% to <10%)) and W#217 (<10%) due to chemical chase is clearly illustrated by the subsequent logs placed in Figure 11. W#542 indicated minor changes in the saturation pre and during the flood, therefore, re-attempts are planned for evaluating the results. The low saturation in the producer W#528 at the onset of the SP indicates the residual condition of the study area (<20% oil saturation). As the flood progresses, and the oil

bank approaches the producer W#528 and therefore increase in oil saturation is observed (upto 40% in the top layer). These results illustrate and confirm the efficiency of the SP flood under field conditions with relatively uniform oil saturation reduction in injector with reasonably good vertical conformance. The saturation logging is limited in the producer due to presence of gravel pack and progressive cavity pump.

Table 2—Water Analysis Pre-flood (Year 2023) and During Flood (Year 2025).

Values	Pre-flood (ppm)	During-flood (ppm)
Salinity	7634	8424
Sulphate ion	19	58
Iron	0.05	0.35
TDS	9750	10470
Calcium ions	104	96
Magnesium ions	53	44

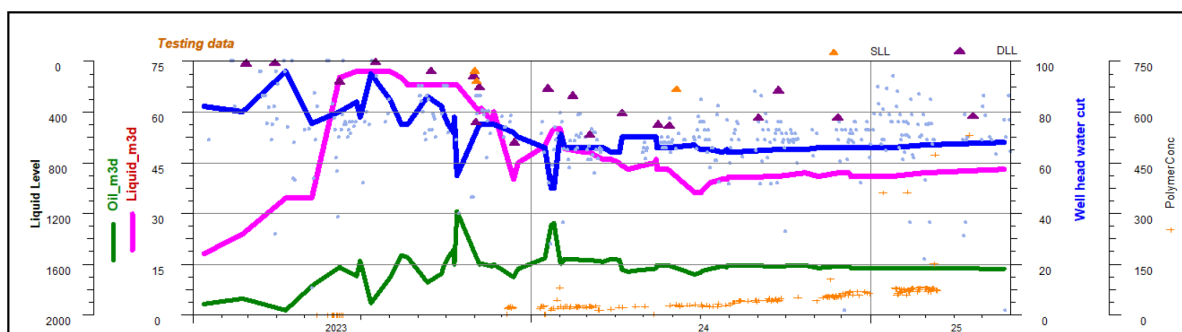


Figure 9—Well Monitoring data (Liquid rate, Oil rate, water-cut (chemistry and monthly average), Dynamic and Static Liquid level, and Polymer Concentration) of W#528 Direct Oil Producer of SP scheme

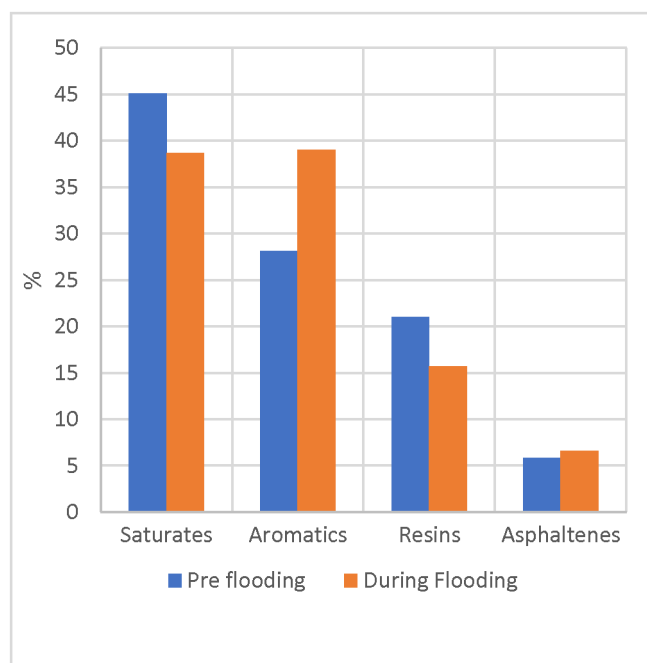


Figure 10—Oil Analysis Pre (Year 2023) and During Flood (Year 2025) in W#528.

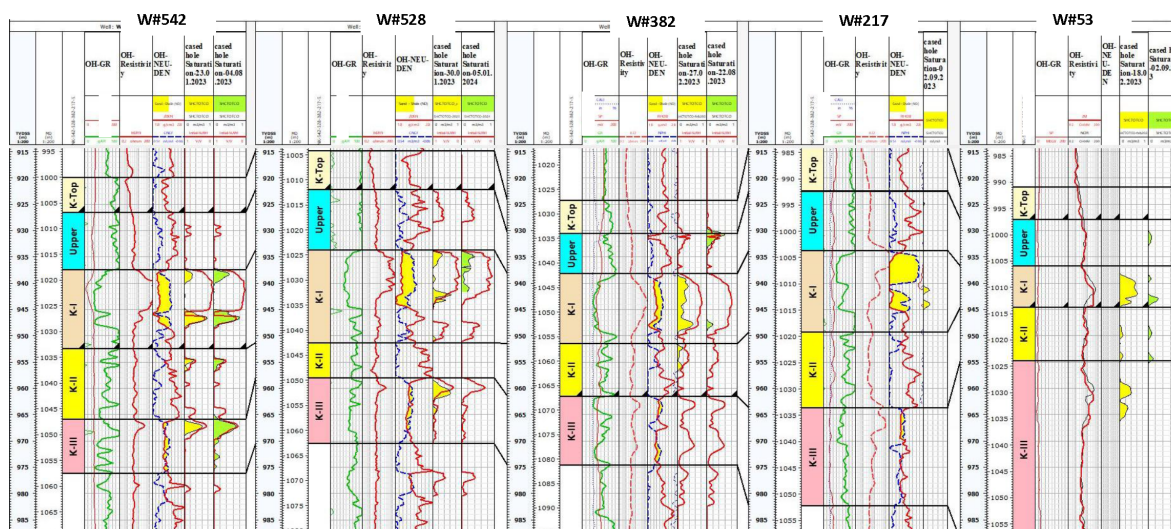


Figure 11—Time Lapse Saturation logging of Pilot wells Pre flood and During Flood for Injectors W#542, W#53, W#382, W#217 and Producer W#528.

Production Performance

The section consists of analysis of polymer-surfactant concentration and water-cut as the SP flooding progresses in the pilot area. The performance analysis is done for the pilot producer and the offset wells of the area. The performance of the study area and pilot producer is placed as Figure 12. Overall, in the study area, as the pilot started the oil was 55 m³/d. As the pilot progressed, the peak oil went up to 80 m³/d. A normal five spot pattern has contributed to a peak incremental oil of 30 m³/d (50% increment in initial oil rate). This increment in oil was due to a decrease in water-cut from 75% to 66% (during peak time) in spite of the increase in liquid rate from 200 m³/d to 290 m³/d. The pilot is currently producing 70 m³/d of incremental oil at 70% WC and 255 m³/d liquid rate. The direct pilot producer W#528 oil production increased from 2.4 m³/d to peak of 21 m³/d (9 times more than base rate). The water-cut declined from 80% to 63% in spite of liquid increase from 25 m³/d to 70 m³/d. The current production rate is 45 m³/d liquid rate, 68% water-cut and 14 m³/d oil rate.

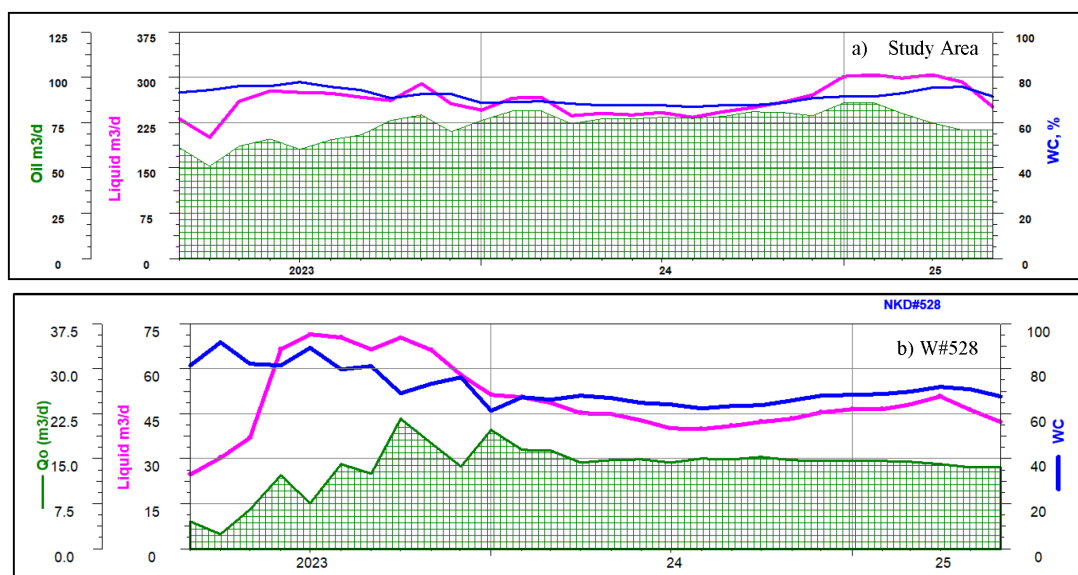


Figure 12—Performance Plot (Oil, Water-cut and Liquid) of a) Study Area b) Pilot producer W#528.

At the onset of the Pilot (March'23)

The water-cut in the pilot producer W#528 was similar to the average water-cut in the study area of 75% as shown in Figure 13. There is high water-cut in the structural lows around producers W#176 and W#326 at the beginning of the pilot. Water-cut was 65% around W#217 as it is most structurally updip well of the fault block. Due to two-way fault closure, the water-cut in the well W#36 was also on the lower side than the average water-cut. W#156 is producing with less water-cut due to execution of the water-shut off and perforation optimization job in late 2021. The average liquid productivity of this area is 20 m³/d and the pilot producer was producing 25 m³/d. In order to confine the response to the pattern area, the liquid rate of the pilot producer was increased to 35 m³/d (based on the availability of artificial lift deliverability rate) at that time.

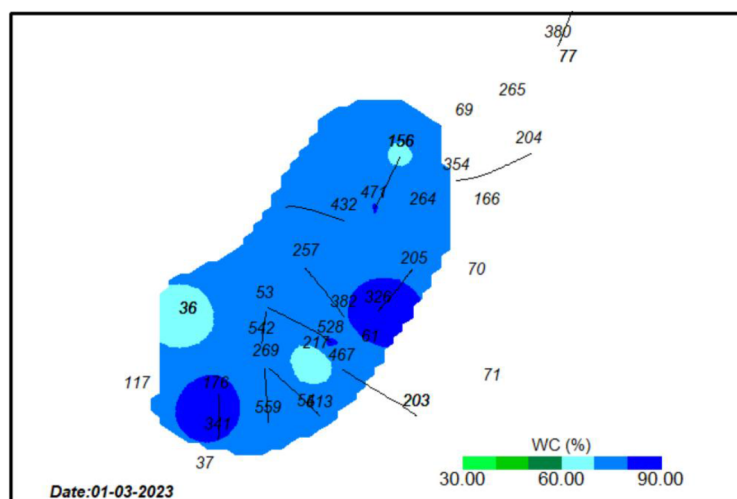


Figure 13—Grid map of Water-cut Pre-Flood (01.03.2023).

At the end of SP Slug (September'23)

The pilot producer W#528 was initially producing 60% of the planned liquid rate due to unavailability of high liquid withdrawal pumps. In June'2023, a high liquid withdrawal pump was installed and the well started producing at the planned liquid rate of 65 m³/d. The rate was based on the 260 m³/d of injection in the pilot area to keep the pattern balanced. Due to the lower liquid withdrawal viz-a-viz planned and historical production-based streamlines, the offset producers W#269 and W#326 witnessed the first response of water-cut decline (as shown in Figure 14). Historically the highest production of the block was from W#70, W#71 (downdip to W#326) and W#37 (downdip to W#269 and W#513). Oil rate of W#269 and W#326 increased to 9 m³/d and 8 m³/d from base rate of 7 m³/d and 3 m³/d respectively during the SP phase. The water-cut decline in W#269 and W#326 was 60% and in W#326 was 30% from initial water-cut of 70% and 80% respectively. The injector-producer distance of W#528 and of the offset wells W#269 and W#326 is the same (~150 m) and average liquid withdrawal of the well was 25 m³/d. The good sweep and vertical conformance in W#382 and W#53 based on saturation logs at the end of the SP slug supports the production performance of W#326. There was no surfactant concentration observed in the producers during this period. At the end of the SP slug, it can be concluded that aquifer movement combined with historical exploitation governs the early response observed in wells W#326 and W#269. Furthermore, liquid withdrawal plays a critical role in displacing the oil front towards the direct oil producer W#528.

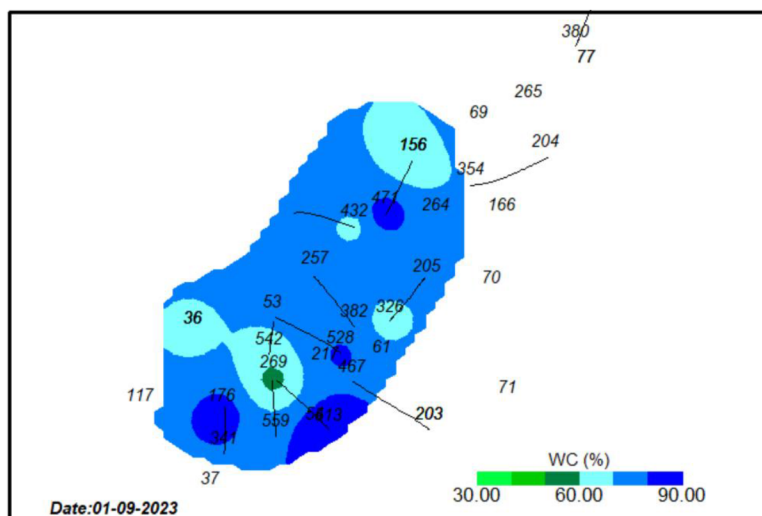


Figure 14—Grip Map of Water-cut at the end of Surfactant-Polymer Slug.

After 6 months of polymer slug/one year of Pilot (March'24)

The pilot producer was producing at the planned rate of 65 m³/d to keep the pattern balanced till October'23. Thereafter, PCP stopped working due to rod unscrew and the well was taken up for intervention. During the intervention, coiled tubing was lowered in the gravel pack for cleaning and huge quantities of produced sand and gravel were recovered. There was a possibility of limited efficiency of gravel pack and sudden failure due to high liquid withdrawal. The well was recompleted with only coiled tubing cleaning of gravel pack as the scope of complete failure was limited as recent gravel pack was installed in January'2023. The gravel pack lifetime is over 10 years in this area. On recompletion within a month, the dynamic liquid level in W#528 dropped to 500 m, and the produced liquid rate dropped to 35 m³/d. Before the drop-in liquid rate, W#528 had good response in terms with water-cut decline to 70 percent and with high liquid withdrawal oil rate increased to 20 m³/d. Even after the liquid drop, well continued to flow at oil rate of 13 m³/d. The first polymer concentration in W#528 was observed in minimal amounts (~20 ppm) in December 2024. These observations clearly indicate that increased liquid withdrawal, coupled with higher injected pore volume, is effectively advancing oil front towards W#528. Due to absence of free water production in both W#326 and W#269, polymer concentration couldn't be ascertained in the produced emulsion-phase water. Thereafter, the well ceased to flow due to gravel pack failure. As W#528 responded to the process, there was considerable production from W#326 and W#269. Gravel pack fishing and recompletion took around 15 non-flowing days. During the workover operation, saturation logging was done, which corroborated the analysis of oil displacement towards W#528 (from residual 20% to 40%). The well was revived in January 2024 and flowed at 50-55 m³/d liquid rates till March 2024. The water cut declined to a minimum of 60% with a peak oil rate of 20 m³/d. Polymer concentrations remained stable to 20-25 ppm (as shown in Figure 15).

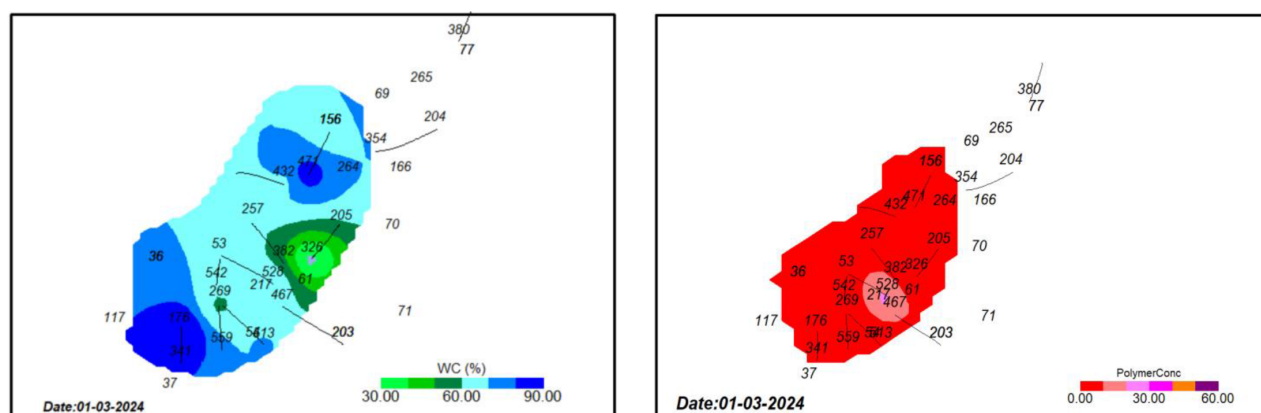


Figure 15—Grid Map of Water-cut and Polymer Concentration after 1 year of Pilot

Current Status (June 2025)

The direct producer W#528 had declining liquid levels since April 2024, and therefore, the productivity of the well declined from 55 m³/d to 40-45 m³/d. Regular polymer concentration monitoring indicated rising concentrations from 20-25 ppm to over 70 ppm. The concentration remained below 100 ppm, so the reason behind the drop in productivity could not be determined till June 2024. It was observed that the bleach test for measurement of produced polymer concentration is sensitive to degree of hydrolysis (only measures amide groups), iron sulfide presence and range limits to 100-250 ppm. Therefore, a laboratory setup was arranged for carrying out starch iodide tests for determination of polymer concentrations. Multiple samples of W#528 were tested, and the results were in the range of 500 ppm (indicating 50% injected concentration breakthrough), causing in the drop of productivity. The starch iodide method is also sensitive to degree of hydrolysis as well (American Petroleum Institute, 1990). Considering the project is operating for only two years and there is direct relation between the degree of hydrolysis and thermal maturation of polymer, so there is low possibility of high degree of hydrolysis of polymer to give erroneous results. Nevertheless, arrangements are in progress for procurement and usage of high-performance liquid chromatography for accurate measurement of polymer (both amide and acrylate groups). Polymer concentration was also observed in offset producers W#471 (40 ppm), W#432 (50 ppm), W#513 (30 ppm), and W#176 (40 ppm). None of these wells have shown any change in water-cut response due to the process. Therefore, these minor concentrations are interpreted as diffusion of polymer in the direction of water-channels. W#326 and W#269 continue to produce at high oil rates as offset producers. As the wells were not producing free water, polymer concentration check was not feasible. The water-cut of the pilot producer W#528 increased from 60 to 68% in the last year with gradual increase in polymer concentration as well. Currently, the well has been produced at a stable oil rate of 14 m³/d for the last two months. The polymer concentration and water-cut mapping in the current scenario is placed as Figure 16. This SP pilot is considered a significant achievement, having been successfully executed in a high-recovery pay sand (42.6%) with a bottom water drive.

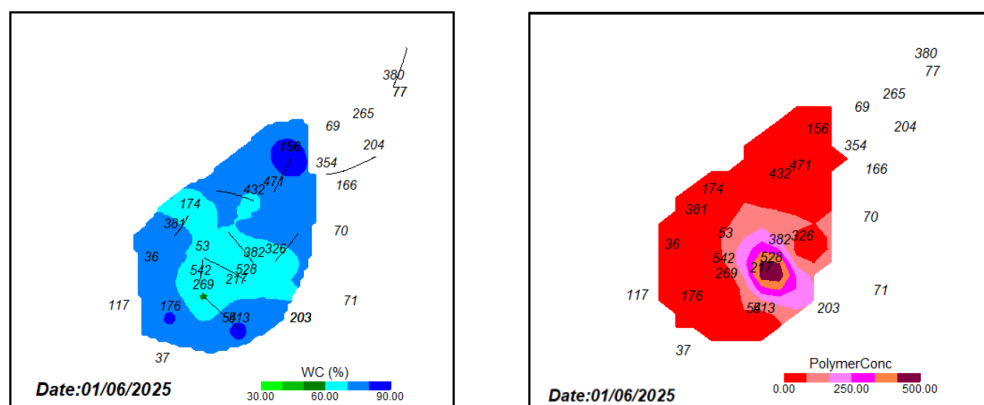


Figure 16—Current Grid Map of Water-cut and Polymer Concentration.

Challenges and Way Forward

- All the data in combination provided a good understanding of the pilot's performance. The enhancement of oil rates pre-flood and during flooding is clearly observed from [Figure 5](#) (pre flooding) and [Figure 17](#) (during flooding). The average oil productivity of pilot area increased from 5 m³/d to over 10 m³/d.
- The next step in the pilot is to design the commercial field scale SP flooding scheme by detailed simulation studies and assess the techno-economic feasibility. The Pilot has already recovered 50% of the investment.
- The pilot would be under evaluation for 9 months to foresee the problems with the high concentration of chemical slug breakthrough. Scale squeeze designs are also being explored to reduce the failure rate of downhole completion and stimulate gravel packs to improve liquid productivity.
- The best methodology to measure produced polymer and surfactant concentration needs further work as surfactant concentration was measured nil and discrepancy in polymer concentration was observed by two different methods. Chromatography, spectroscopy, magnetic resonance, titration etc. are being assessed for a comparative study.
- Finally, quality assurance of the chemicals in the project is to be improved upon by inspection of all chemicals and molecular level fingerprinting.

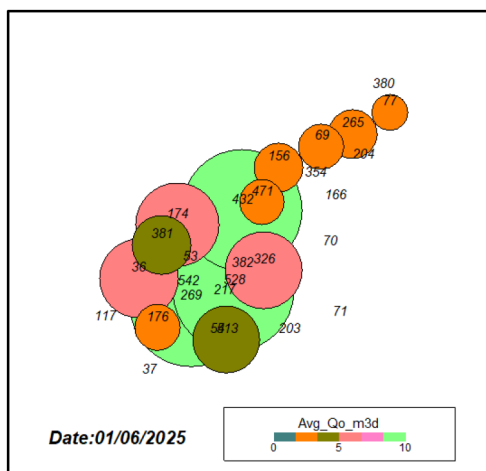


Figure 17—Bubble Map of Current oil rate

Conclusion

The main conclusions drawn from pilot operations are the following:

- I. Injection of good quality chemical slugs and stringent quality monitoring is the key to the success of the SP flood.
- II. SP Slug reduced water-cut in direct as well as offset producers of the pattern. Therefore, the process works successfully with high pay thickness (~10 m) and strong water drive.
- III. The closed normal five spot system effectively tested the efficacy of the process in two years' time.
- IV. The chemical front movement is mainly governed by historical exploitation, gravity and aquifer entry. Therefore, the response of the chemical slug not only remains in the patterns but offset wells also saw an increase in oil production.
- V. Saturation logging indicated good vertical conformance, ROS mobilization from the injectors and an increase in oil saturation towards the producer. Thus, SP slug targeted the mobile and residual oil in the system.
- VI. As the flood progresses, failure of the completion equipment and drop in liquid productivity might be aggravated by high withdrawal rates, scaling issues, changes in oil characteristics and increase in chemical slug concentrations in production fluids.
- VII. Measurement methods of chemical slugs in the produced fluids need to be assessed on case-to-case basis.

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Abbreviations

SP	Surfactant Polymer
ASP	Alkali Surfactant Polymer
ROS	Residual Oil Saturation
ppm	parts per million
ppb	parts per billion
PCP	Progressive Cavity Pumps
QA/QC	Quality Assurance/Quality Check
SATG	Saturation Gas
MM	Million
STOIIP	Stock tank oil in place
IFT	Interfacial Tension
TDS	Total Dissolved Solids
SLL	Static Liquid Level
DLL	Dynamic Liquid Level
C/O	Carbon Oxygen Bond.
CMC	Critical Micelle Concentration
SARA	Saturates, Aromatics, Resins, Asphaltenes
WC	Water-Cut
PV	Pore Volume

NTU	Nephelometric Turbidity unit
pH	Potential of Hydrogen
cP	Centipoise
SRB	Sulphate reducing bacteria
TSS	Total Suspended Solids
NaCl	Sodium Chloride
OWC	Oil Water Contact
GOC	Gas Oil Contact
Qo	Oil rate
Np	Cumulative Production
EOR	Enhanced Oil Recovery

References

- Alshaikh, M., Lee, Y. S. & Hascakir, B., 2019. *Anionic Surfactant and Heavy Oil Interaction during Surfactant-Steam Process*. San Jose, Society of Petroleum Engineers.
- Al-Shuaili, K. et al, 2018. *Field Piloting of Alkaline Surfactant Polymer in Sultanate of Oman*. Muscat, Society of Petroleum Engineers.
- American Petroleum Institute, 1990. *Recommended Practices for Evaluation of Polymers Used in Enhanced Oil Recovery Operations*. 1st ed. Washington: REG. US. Patent Office.
- Chang, H. L., Guo, Z. & Zhang, J., 2024. *Polymer Flooding: The Good, the Bad, and the Ugly - Lessons Learned from Field Practices*. Tulsa, Society of Petroleum Engineers.
- Jain, A. K., Dhawan, A. K. & Misra, T. R., 2012. *ASP Flood Pilot in Jhalora (K-IV) - A Case Study*. Mumbai, Society of Petroleum Engineers.
- Olajire, A. A., 2014. Review of ASP EOR (alkaline surfactant polymer enhanced oil recovery) technology in the petroleum industry: Prospects and challenges. *Energy*, **77**(1), pp. 963-982.
- Pandey, A., Koduru, N., Stanley, M. & Pope, G. A., 2016. *Results of ASP Pilot in Mangala Field: A Success Story*. Tulsa, Society of Petroleum Engineers.
- Shankar, V. et al, 2022. Mangala Polymer Flood Performance: Connecting the Dots Through In-Situ Polymer Sampling. *SPE Reservoir Evaluation & Engineering*, **25**(04), p. 655–666.
- Sheng, J. J., 2011. Surfactant Flooding. In: E. Inc., ed. *Modern Chemical Enhanced Oil Recovery: Theory and Practice*. January 2010. Authors: James J. Sheng. Lubbock: Elsevier Inc., pp. 239-335.
- Singh, B. P., Parulkar, S. & Kumar, A., 2017. *Successful Pilot Implementation of ASP Flooding through CEOR/IOR Application – A Case Study*. Mumbai, Society of Petroleum Engineers.
- Thomas, A., 2019. *Essentials of Polymer Flooding Technique*. First ed. New Jersey: John Wiley & Sons.
- Watson, A., Trahan, G. A. & Sorensen, W., 2014. *An Interim Case Study of an Alkaline-Surfactant-Polymer Flood in the Mooney Field, Alberta, Canada*. Tulsa, Society of Petroleum Engineers.
- Zagitov, R., Mathur, V., Subramanian, J. & Venkat, P. S., 2020. *Investigation on Fouling During Application of EOR Polymer in Aishwarya Field*. Abu Dhabi, Society of Petroleum Engineers, pp. 1-20.